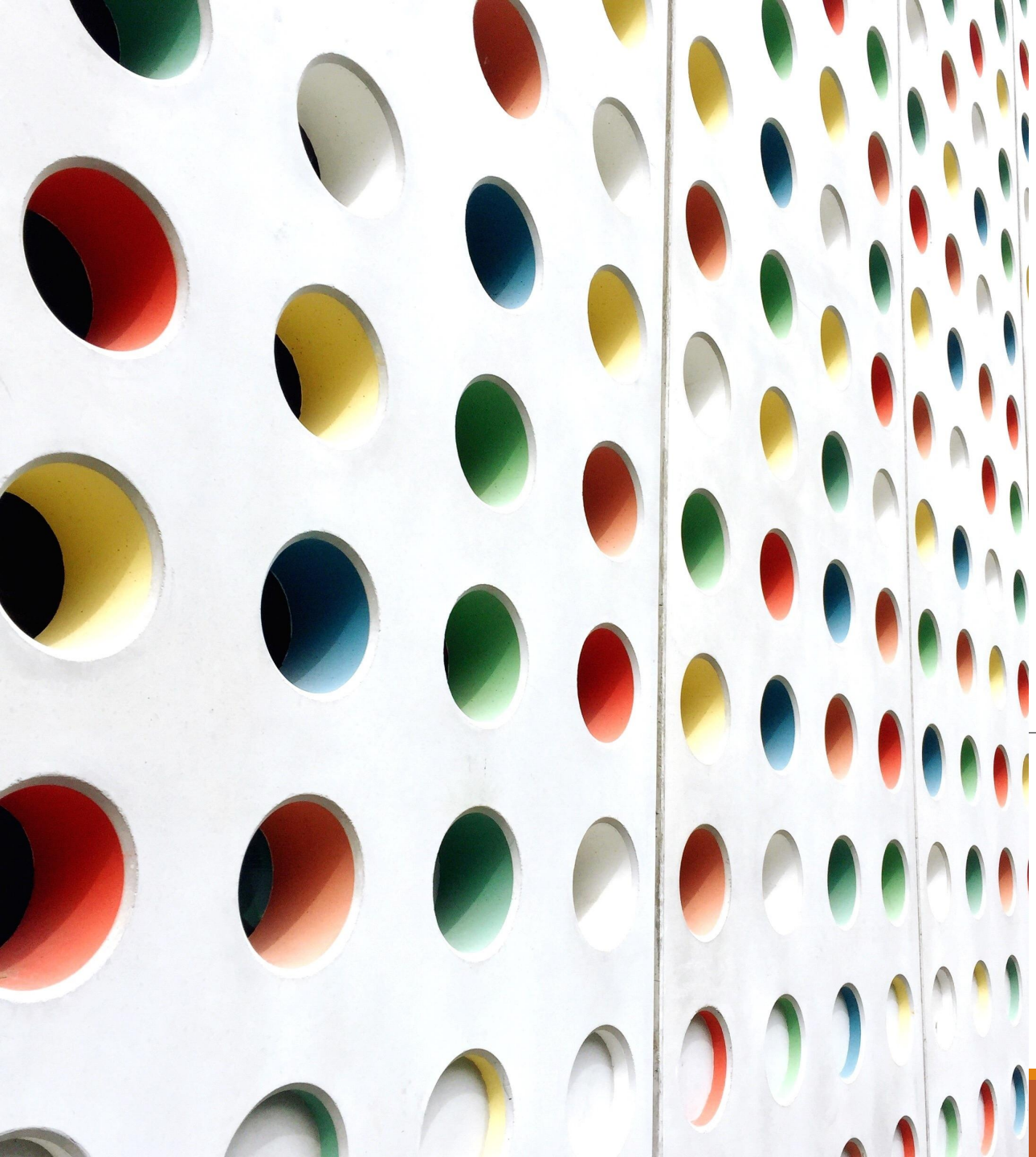


Graph

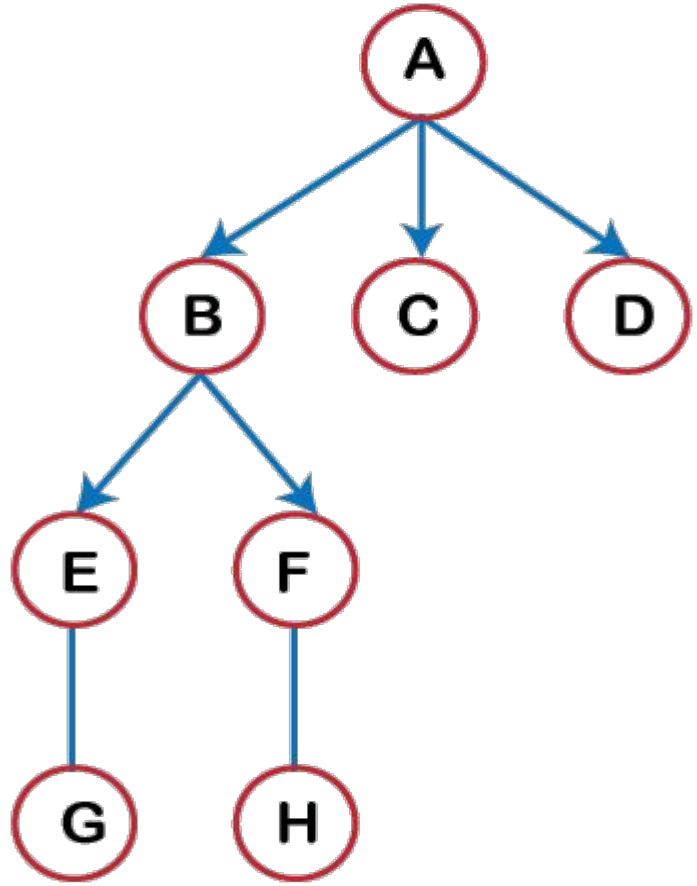
M. ASAD ARSHED

Graph

A graph in C++ is a non-linear data structure defined as a collection of vertices and edges. Linear data structure is a structure in which all the elements are stored sequentially and have only single level. In contrast, a non-linear data structure is a structure that follows a hierarchy, i.e., elements are arranged in multiple levels.

What is Non-Linear?

In the above figure, we can assume the company hierarchy where A represents the CEO of the company, B, C and D represent the managers of the company, E and F represent the team leaders, and G and H represent the team members. This type of structure has more than one level, so it is known as a non-linear data structure.



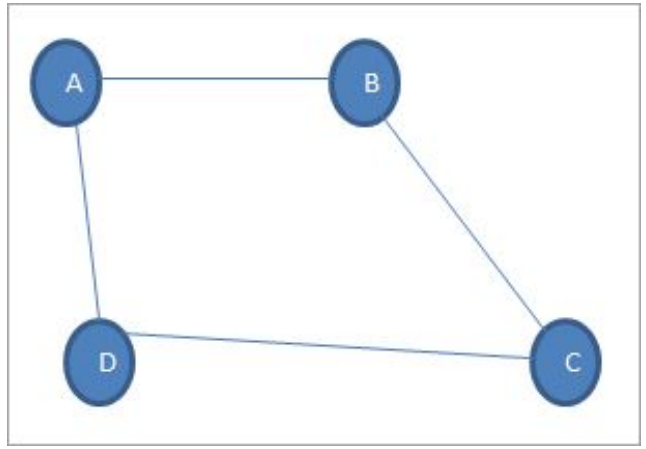
Graph vs Tree

The basis of Comparison	Graph	Tree
Definition	Graph is a non-linear data structure.	Tree is a non-linear data structure.
Structure	It is a collection of vertices/nodes and edges.	It is a collection of nodes and edges.
Edges	Each node can have any number of edges.	If there is n nodes then there would be n-1 number of edges
Types of Edges	They can be directed or undirected	They are always directed
Root node	There is no unique node called root in graph.	There is a unique node called root(parent) node in trees.
Loop Formation	A cycle can be formed.	There will not be any cycle.
Traversal	For graph traversal, we use Breadth-First Search (BFS), and Depth-First Search (DFS).	We traverse a tree using in-order, pre-order, or post-order traversal methods.
Applications	For finding shortest path in networking graph is used.	For game trees, decision trees, the tree is used.

Terminologies

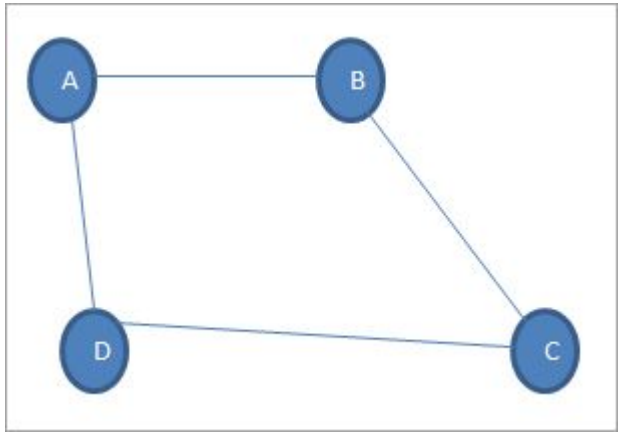
Vertex: Each node of the graph is called a vertex. In the above graph, A, B, C, and D are the vertices of the graph.

Edge: The link or path between two vertices is called an edge. It connects two or more vertices. The different edges in the above graph are AB, BC, AD, and DC.



Terminologies

Adjacent node: In a graph, if two nodes are connected by an edge then they are called adjacent nodes or neighbors. In the above graph, vertices A and B are connected by edge AB. Thus A and B are adjacent nodes.



Terminologies

Closed path: If the initial node is the same as a terminal node, then that path is termed as the closed path.

Simple path: A closed path in which all the other nodes are distinct is called a simple path.

Cycle: A path in which there are no repeated edges or vertices and the first and last vertices are the same is called a cycle. In the above graph, A->B->C->D->A is a cycle.

Terminologies

Connected Graph: A connected graph is the one in which there is a path between each of the vertices. This means that there is not a single vertex which is isolated or without a connecting edge. The graph shown above is a connected graph.

Complete Graph: A graph in which each node is connected to another is called the Complete graph. If N is the total number of nodes in a graph then the complete graph contains N(N-1)/2 number of edges.

Terminologies

Weighted graph: A positive value assigned to each edge indicating its length (distance between the vertices connected by an edge) is called weight. The graph containing weighted edges is called a weighted graph. The weight of an edge e is denoted by w(e) and it indicates the cost of traversing an edge.

Diagraph: A digraph is a graph in which every edge is associated with a specific direction and the traversal can be done in specified direction only.

Graph Representation

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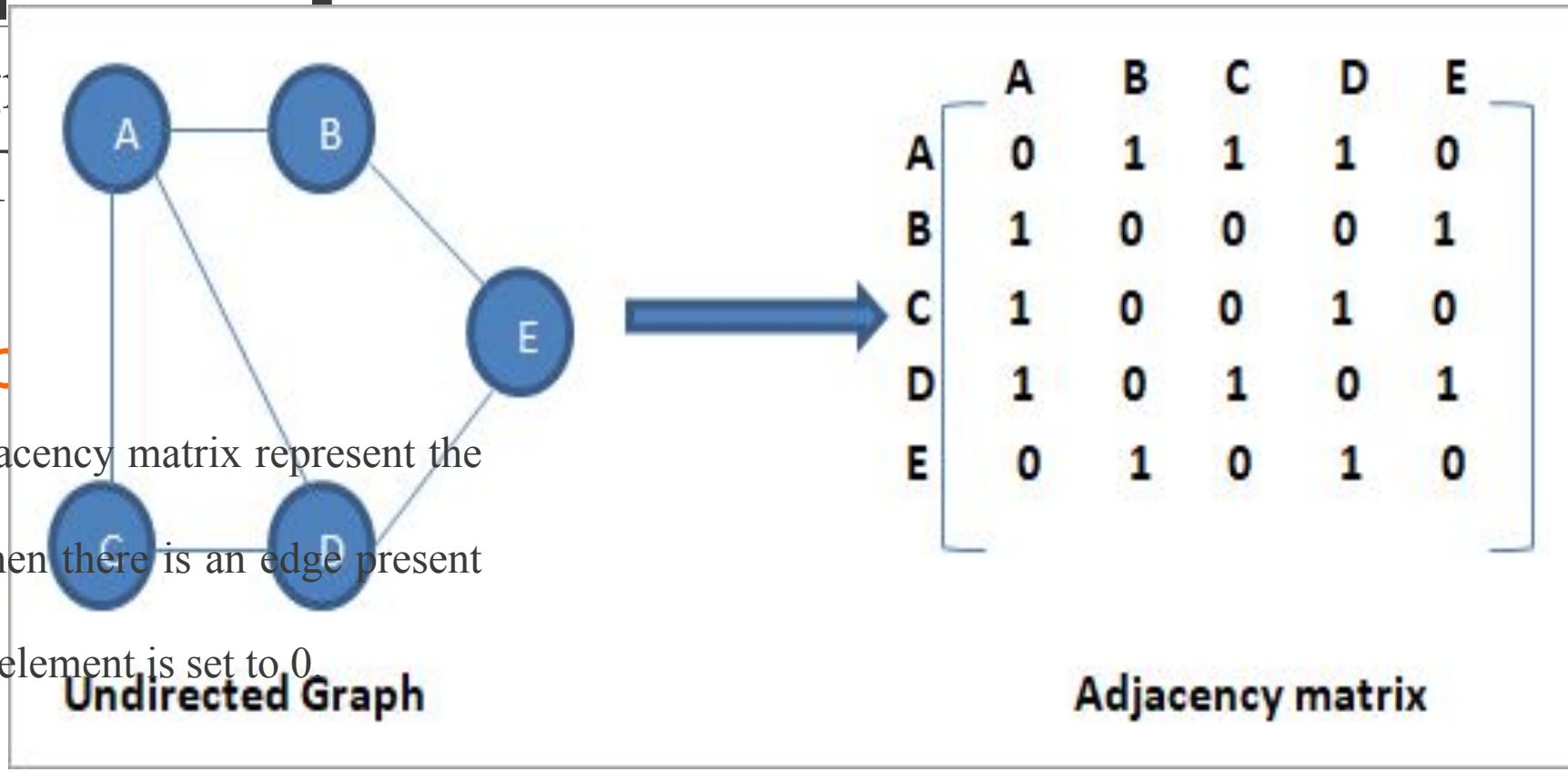
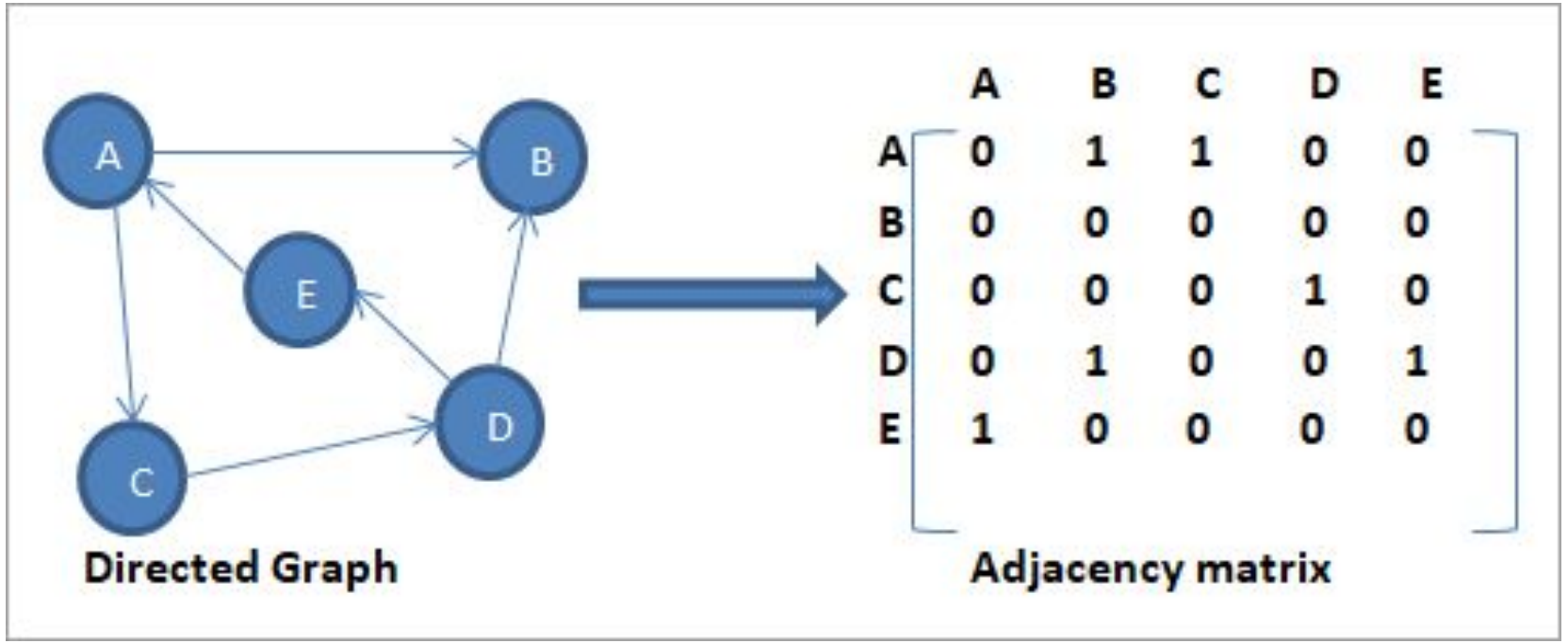
1. Sequential Rep

- The rows and columns of the adjacency matrix represent the vertices in a graph.
- The matrix element is set to 1 when there is an edge present between the vertices.
- If the edge is not present then the element is set to 0



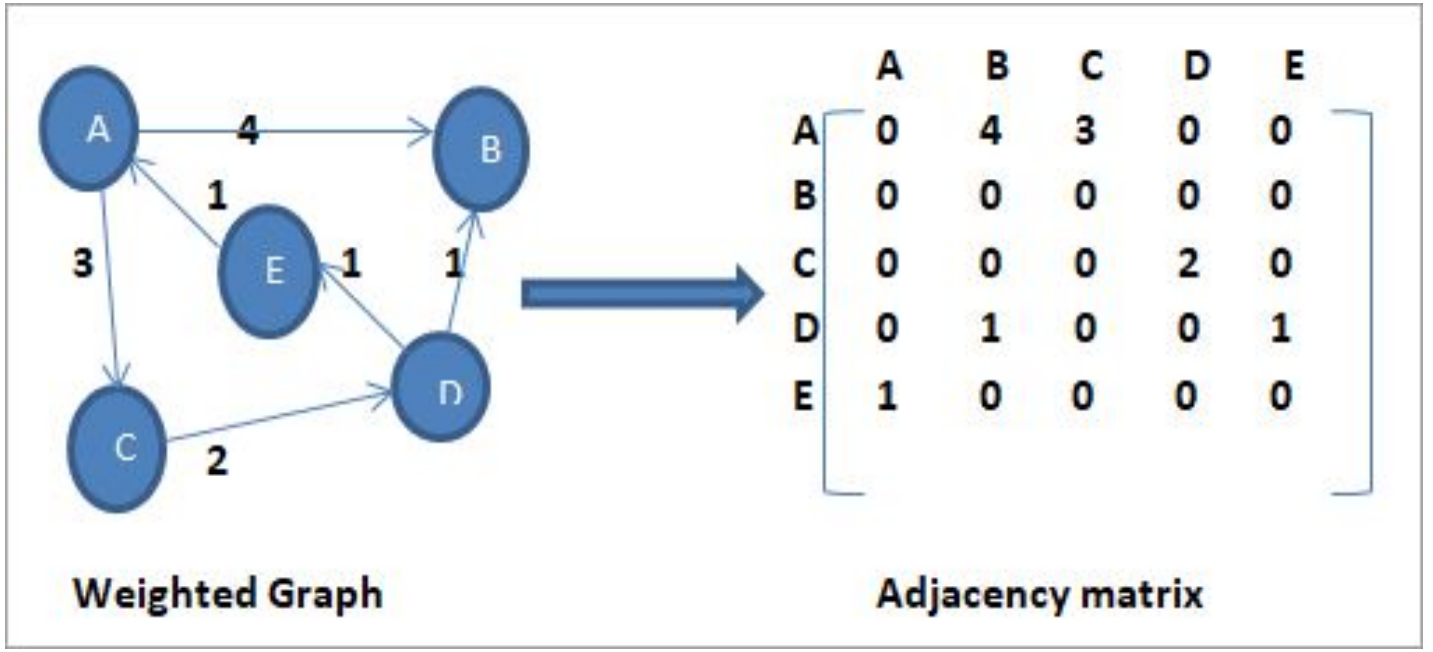
Graph Representation

Adjacency matrix of a Directed Graph.



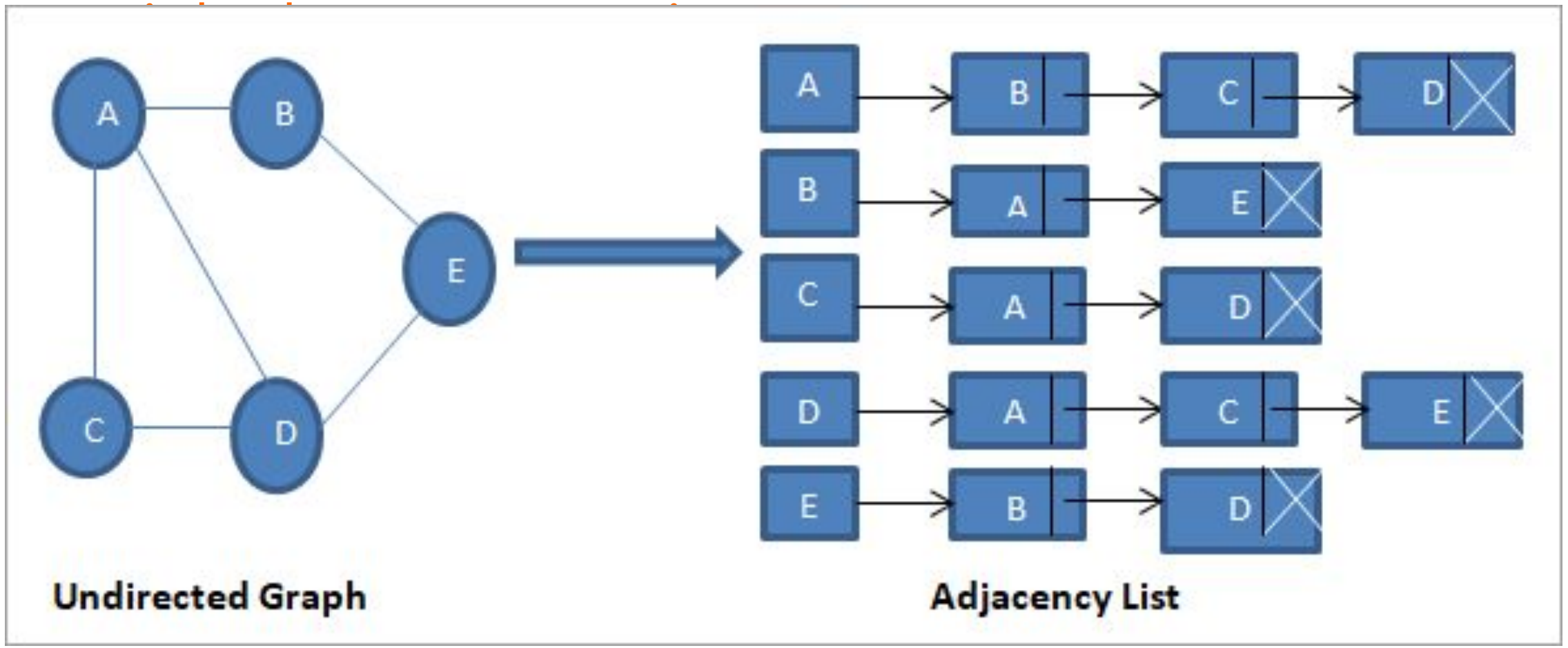
Graph Representation

weighted graph and its corresponding adjacency matrix.

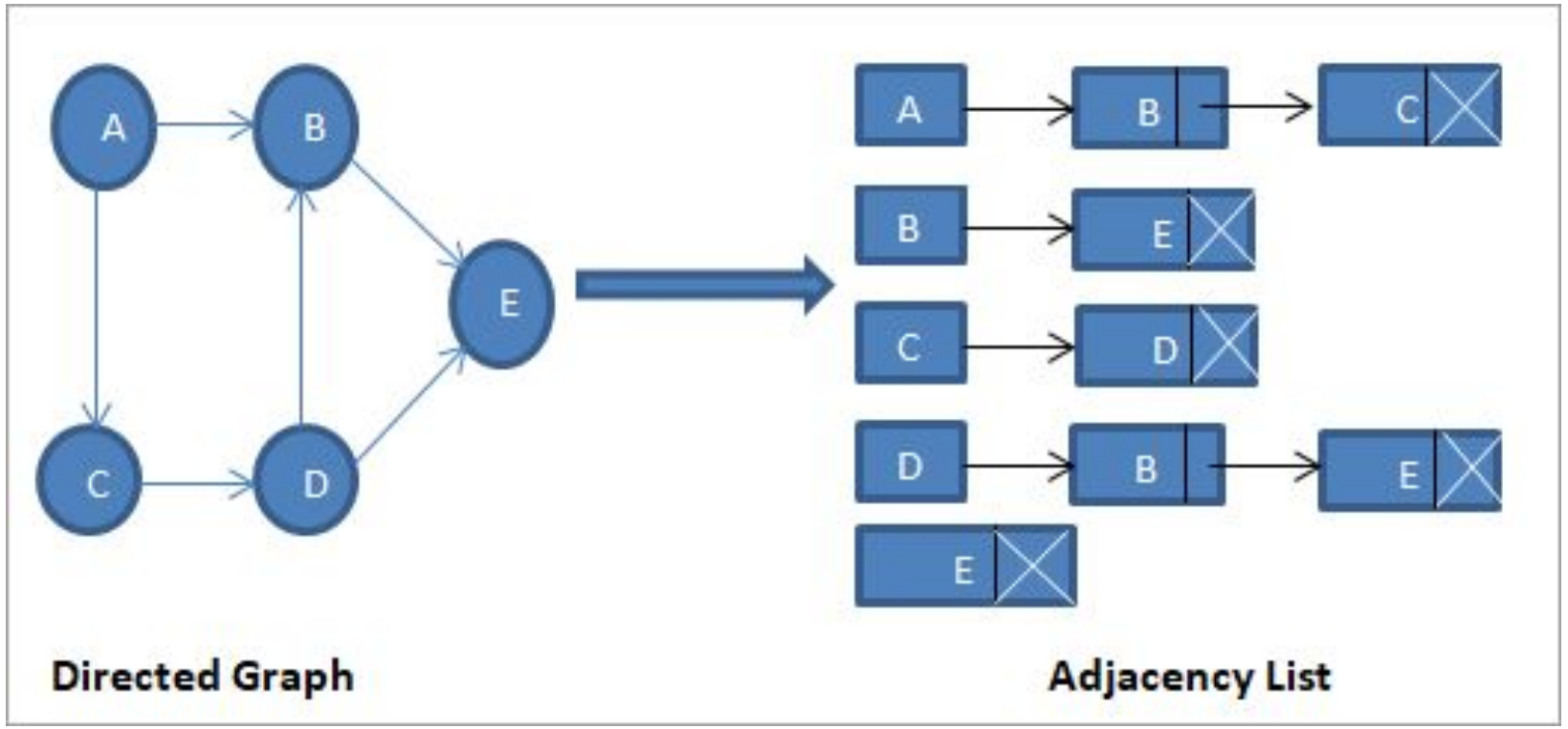


	A	B	C	D	E
A	0	4	3	0	0
B	0	0	0	0	0
C	0	0	0	2	0
D	0	1	0	0	1
E	1	0	0	0	0

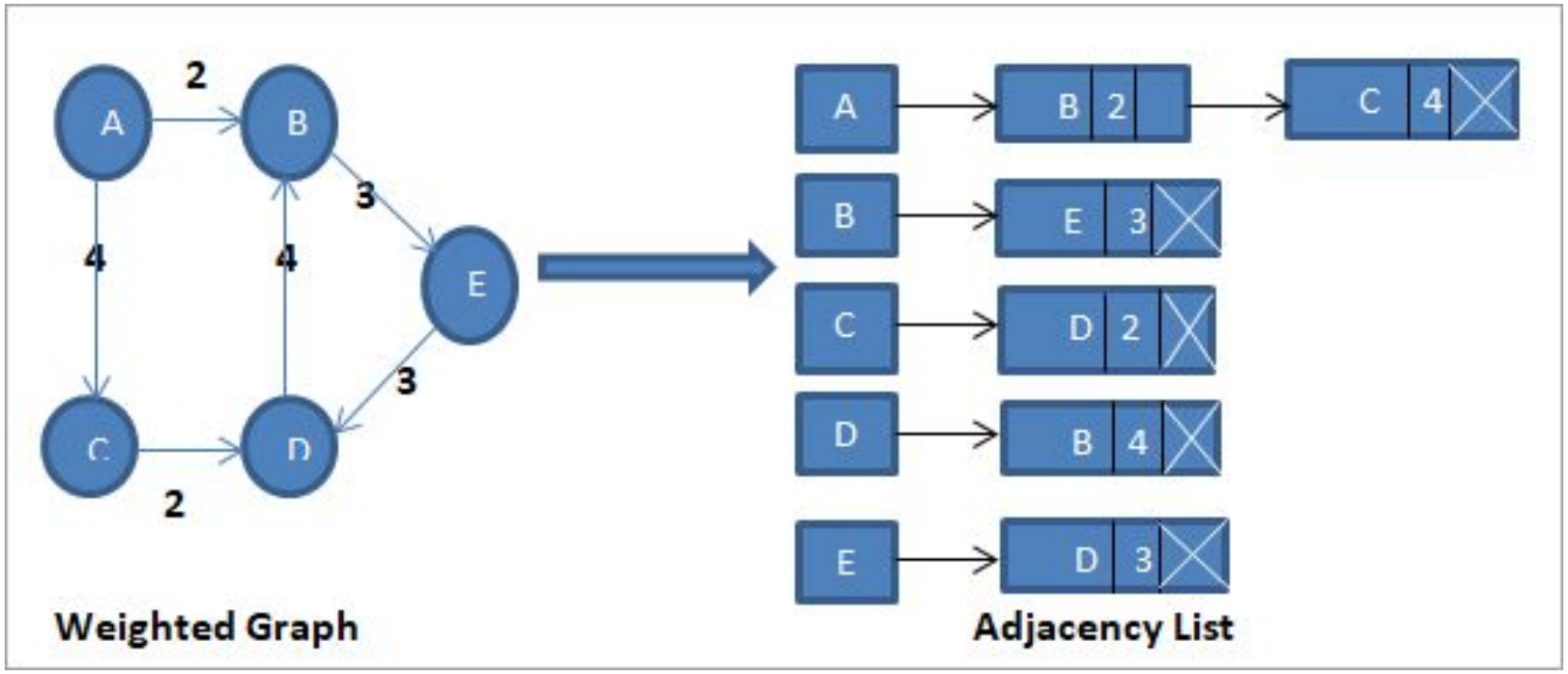
Graph Representation



Graph Representation



Graph Representation



Basic Operations

Add a vertex: Adds vertex to the graph.
Add an edge: Adds an edge between the two vertices of a graph.
Display the graph vertices: Display the vertices of a graph (Traversing)